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No.

ODD END SEMESTER EXAMINATION DECEMBER - 2025

Name of the Course: B.Tech Aerospace Engineering

Semester: 3rd

Name of the Paper: Applied Mechanics

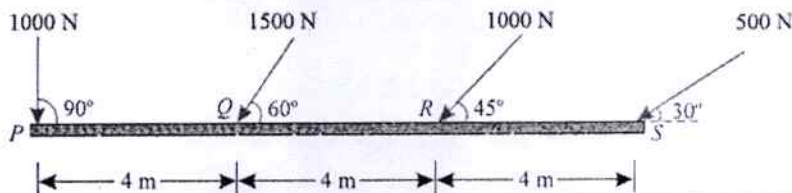
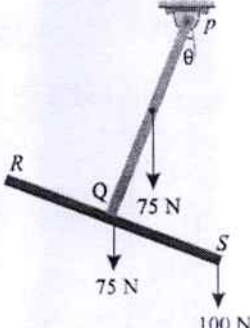
Paper Code: TAS305

Time: 3 Hours

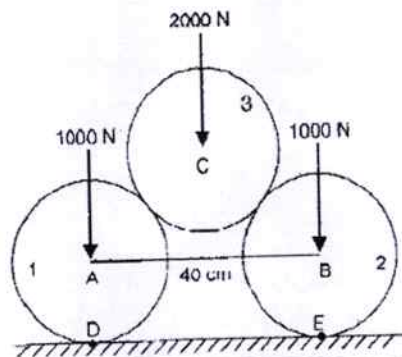
Maximum Marks: 100

Note:

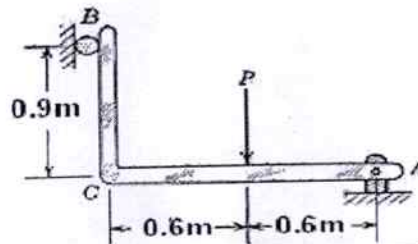
- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question.
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks

Q1	(20 marks)	
(a)	i. Define the term 'force', and state clearly the effects of force. [3 Mark] ii. Distinguish clearly between resolution of forces and composition of forces. [3 Marks] iii. State triangle law of forces and polygon law of forces. [4 Marks]	
(b)	A horizontal line PQRS is 12 m long, where $PQ = QR = RS = 4$ m. Forces of 1000 N, 1500 N, 1000 N and 500 N act at P, Q, R and S respectively with downward direction. The lines of action of these forces make angles of 90° , 60° , 45° and 30° respectively with PS. Find the magnitude, direction and position of the resultant force. 	CO1
(c)	Two identical prismatic bars PQ and RS each weighing 75 N are welded together to form a Tee and are suspended in a vertical plane as shown in Figure below. Calculate the value of θ , that the bar PQ will make with vertical when a load of 100 N is applied at S. Assume length of each bar is 1 meter. 	
Q2	(20 marks)	
(a)	i. What is meant by moment of a force? How will you explain it mathematically? [5 Marks] ii. What is meant by equilibrium? State the conditions of static equilibrium for both coplanar concurrent and non-concurrent force system. [5 Marks]	CO2

- (b) Two smooth circular cylinders, each of weight $W = 1000\text{ N}$ and radius 15 cm , connected at their centres by a string AB of length 40 cm and rest upon a horizontal plane supporting above them a third cylinder of weight 2000 N and radius 15 cm as shown in fig. below. Find the force in string AB and the pressure produced on the floor at the points of contact D and E .



- (c) Find the reactions R_a and R_b induced at the supports A and B of the right angle bar ACB supported as shown in Figure below and subjected to a vertical load P applied the mid-point of AC .

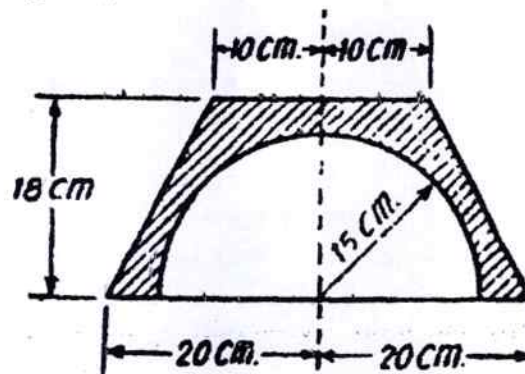


Q3

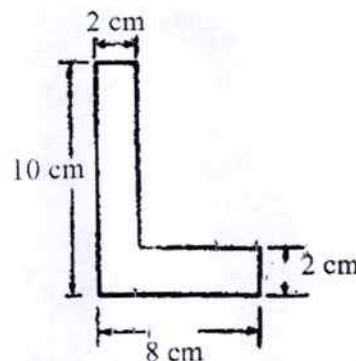
(20 marks)

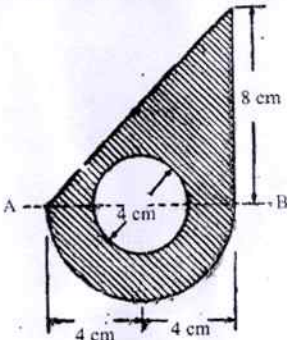
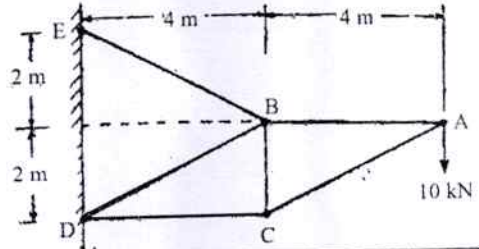
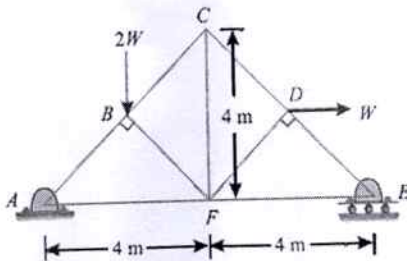
CO3

- (a) Determine the center of gravity of the plane figure as shown below.



- (b) Determine the moment of inertia about centroidal axes XX and YY of the section shown below.



(c)	Determine the moment of Inertia of the shaded area about the axis AB. 	
Q4	(20 marks)	
(a)	Differentiate between determinate and indeterminate structures. Mention the boundary conditions for following support conditions: Simple support, Hinged support, fixed support and free support along with their practical applications.	CO4
(b)	Determine the forces in each member of pin-jointed frame as shown below using method of joint. The truss is hinged at D and E and loaded at A. 	
(c)	Determine the forces in each member of the truss as shown below by using method of sections. Assume $W = 10$ kN. 	
Q5	(20 marks)	
(a)	What do you understand by the term 'acceleration'? Define positive acceleration and negative acceleration. What is the difference between uniform acceleration and variable acceleration?	CO5
(b)	The equation of motion of a particle moving in a straight line is given by : $s = 18t + 3t^2 - 2t^3$ where (s) is in metres and (t) in seconds. Find (i) velocity and acceleration at start, (ii) time, when the particle reaches its maximum velocity, and (iii) maximum velocity of the particle.	
(c)	An aeroplane is flying on a straight level course at 200 km per hour at a height of 1000 metres above the ground. An anti-aircraft gun located on the ground fires a shell with an initial velocity of 300 m/s, at the instant when the plane is vertically above it. At what inclination, to the horizontal, should the gun be fired to hit the plane? What time after firing, the gun shell will hit the plane? What will then be the horizontal distance of the plane from the gun?	

